Automatic Colorization

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Some results of our predictions

Image Color Space

RGB

- an additive color model which added red, green and blue light
- for the sensing, representation and display of images in electronic systems
- used in conventional photography.

YUV

- a color space typically used as part of a color image pipeline
- luma component already existed as the black and white signal
- the UV signal is able to add color

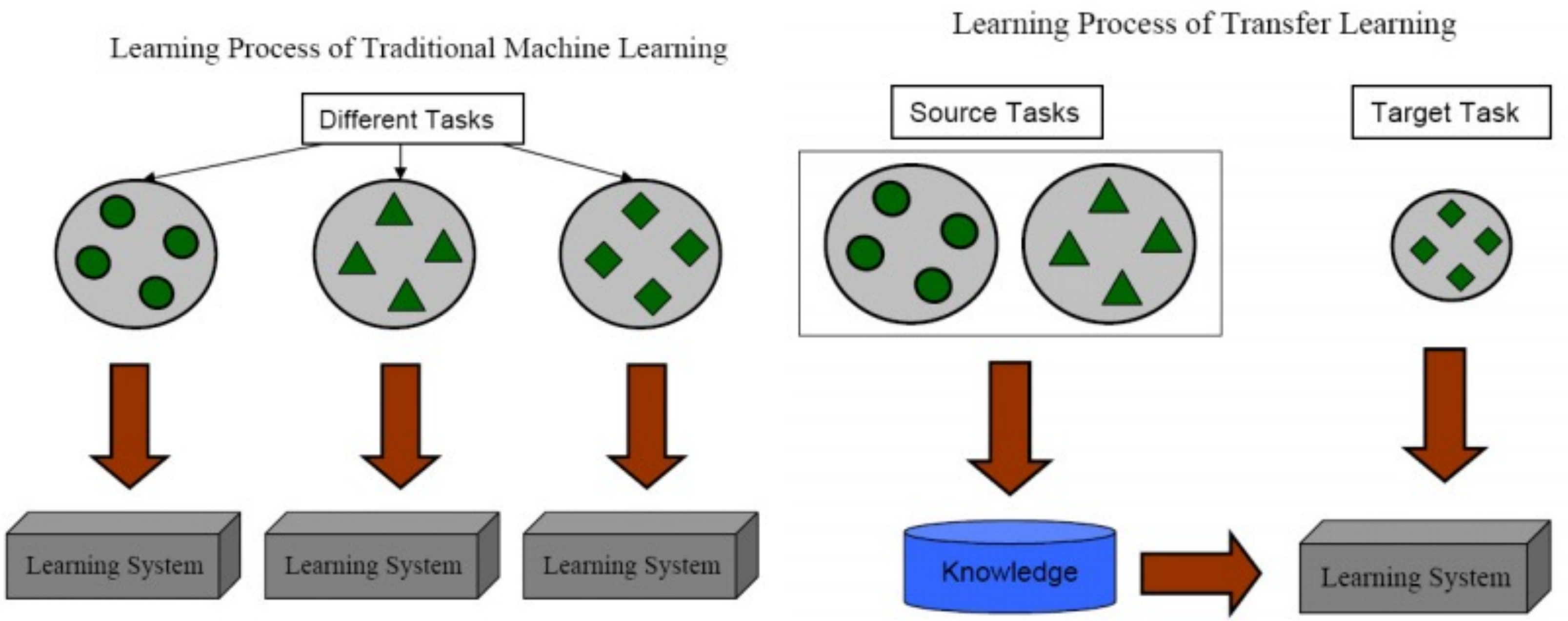
RGB to YUV

$$\begin{bmatrix} Y \\ U \\ V \end{bmatrix} = \begin{bmatrix} 0.299 & -0.169 & 0.499 \\ 0.587 & -0.331 & -0.418 \\ 0.114 & 0.499 & -0.0813 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Transfer Learning

- Transfer learning or inductive transfer is a research problem in machine learning that focuses on storing knowledge gained while solving one problem and applying it to a different but related problem. For example, knowledge gained while learning to recognize cars could apply when trying to recognize trucks. This area of research bears some relation to the long history of psychological literature on transfer of learning, although formal ties between the two fields are limited.

Transfer Learning

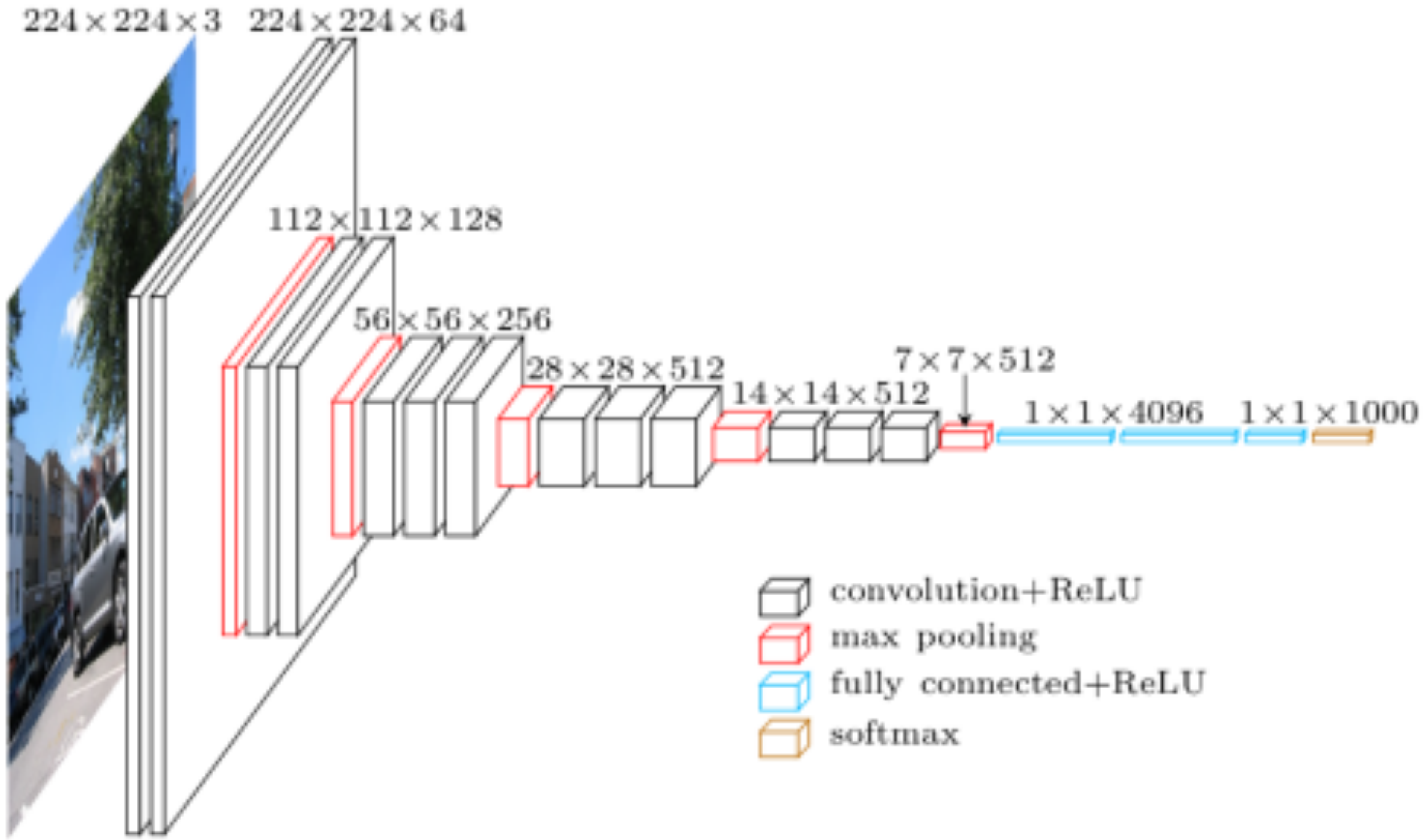


(a) Traditional Machine Learning (b) Transfer Learning

- In CNN classification models, more information can be extracted than just the final classification. It turns out that objects like car wheels and people already start becoming identifiable by layer three. Intermediate layers in classification models can provide useful color information.

- Here we use a pre-trained image classification model to extract features for colorization. We choose the VGG-16 model because it has a simple architecture yet still competitive.

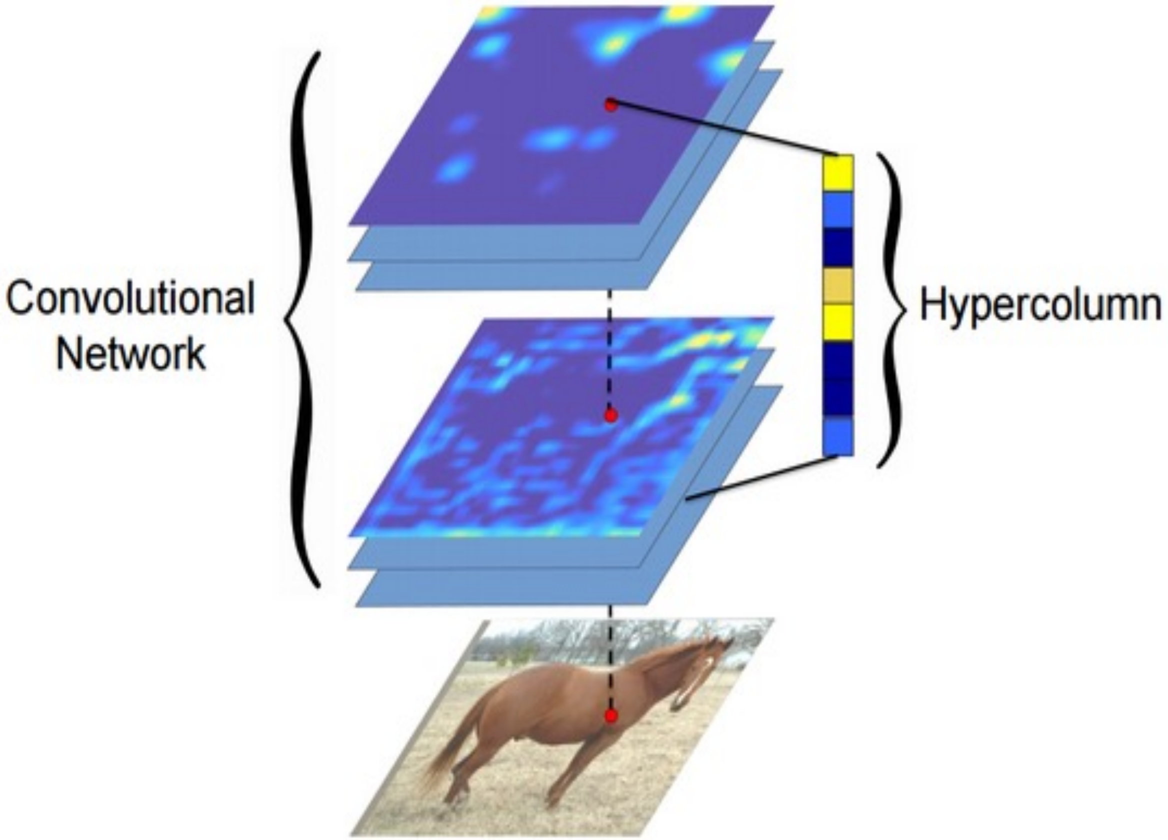
Structure of Vgg16



Hyper Columns

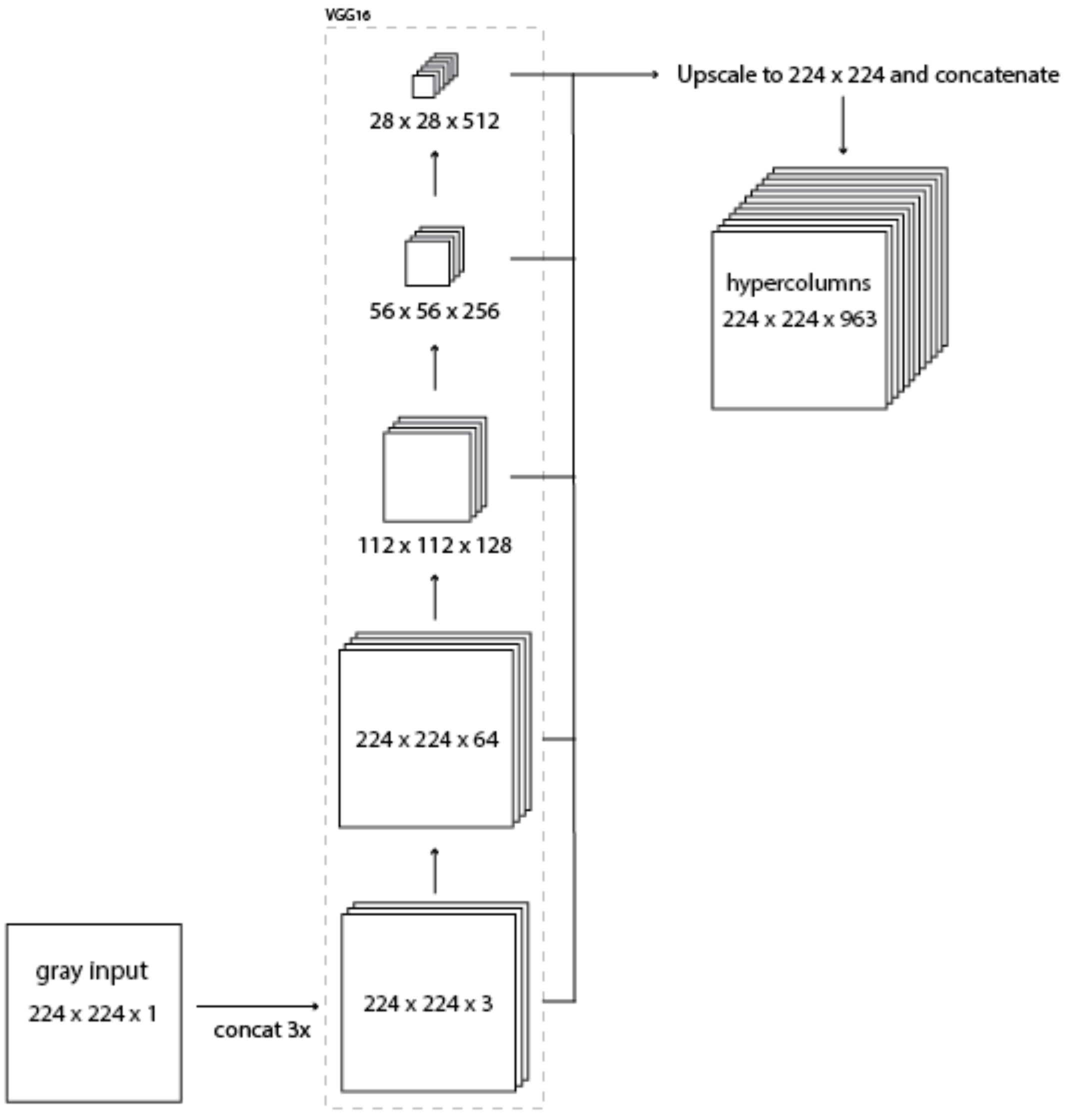
- A hyper column for a pixel in the input image is a vector of all the activations above that pixel. The way I implemented this was by forwarding an image thru the VGG network and then extracting a few layers, upscaling them to the original image size, and concatenating them all together.

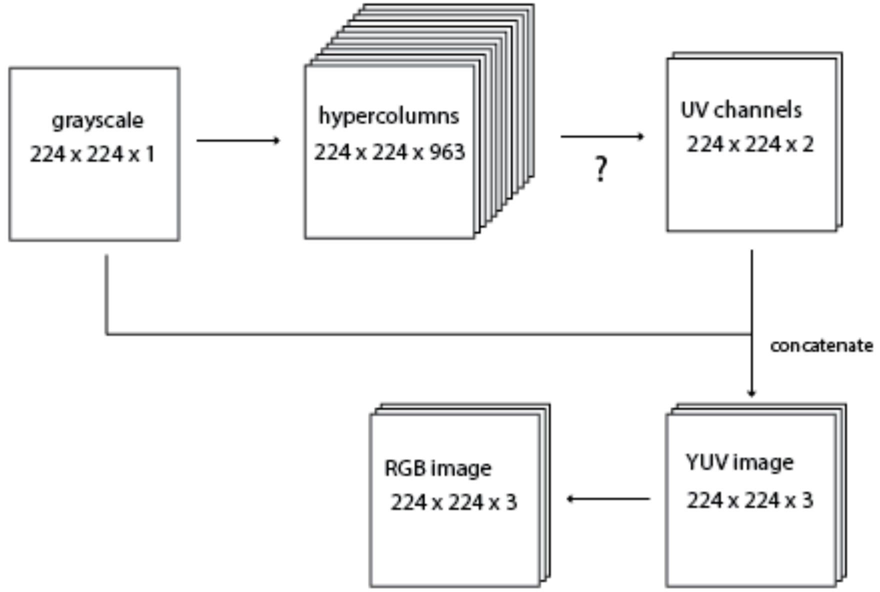
Hyper Columns



Structure of Our Model

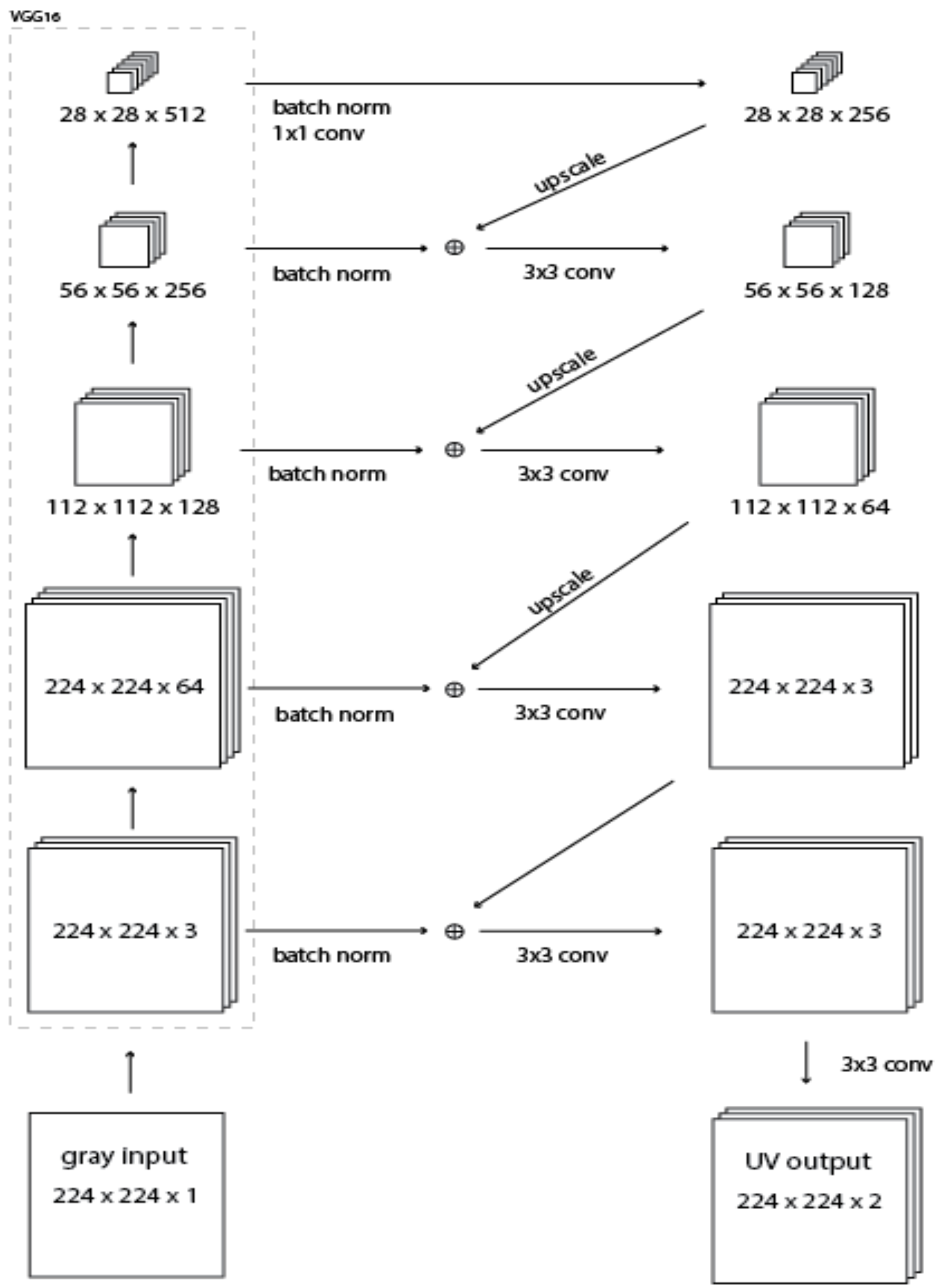
--Hyper Columns





Structure of Our Model

--residual encoder



Loss Function

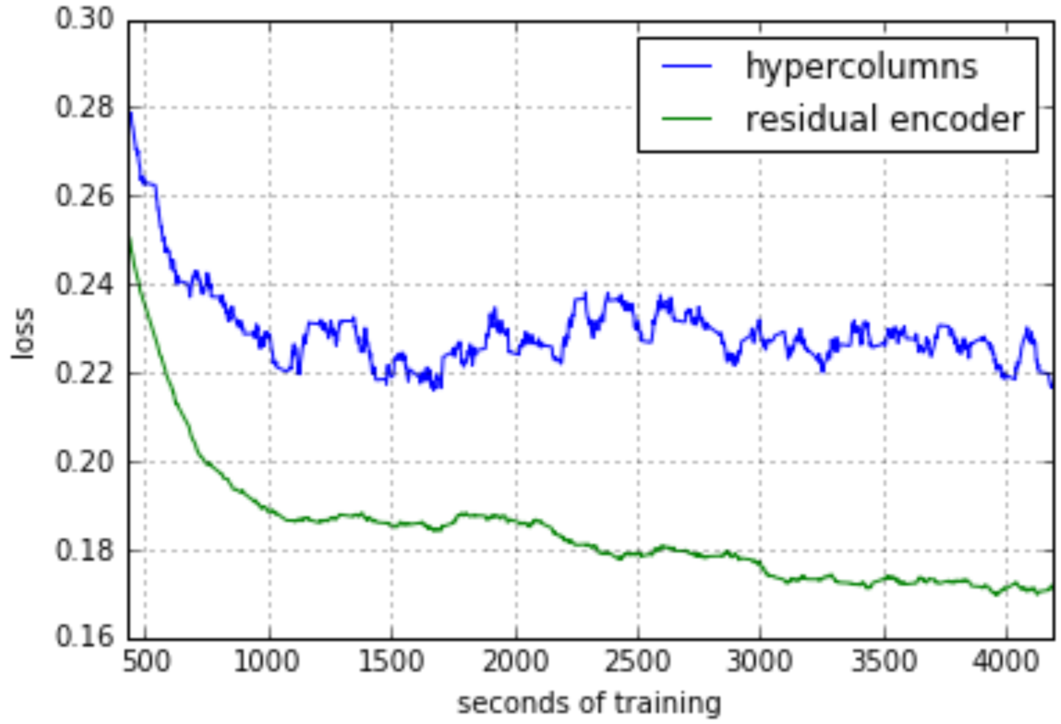
$$loss = [(Pre_{i,j} - Ori_{i,j})^2]_{i \in [0,224], j \in [0,224]}$$

Image Data Set

- Pictures of objects belonging to 101 categories. About 40 to 800 images per category. Most categories have about 50 images. Collected in September 2003 by Fei-Fei Li, Marco Andreetto, and Marc 'Aurelio Ranzato. The size of each image is roughly 300 x 200 pixels.

- Tom and Jerry 166

Training Loss



Predictions

Hyper Columns

Residual Encode

Our Predictions

psnr: 39.492
ssim: 0.915

psnr:39.708
ssim: 0.969

Our Predictions

psnr: 39.899
ssim: 0.976

psnr： 40.053
ssim： 0.994

Let’s enjoy a movie

Outlook

- ➢Model: VGG-19，ResNet
- ➢Loss Function: 1/psnr, others
- ➢Picture: picture size

THANKS